

relying heavily on data, we are forging a synergy between neuroscience and AI, termed as the 'Nature-inspired Framework For Driving.' This framework emulates the human brain's ability to learn and adapt without inundating it with vast amounts of data, starting with a basic understanding of the environment and evolving from there. Thus NIA-based models can create self-driving cars that take actions similar to humans and learn from their experiences to adapt to new situations better.

Doesn't the human brain also rely on data to make decisions?

Traditional algorithms often employ pixel-to-pattern matching techniques for decision-making. Due to which they lack the basic understanding of the environment surrounding them and often make bad decisions. For instance, in object detection, algorithms analyse pixel distributions based on the labelled data. In contrast, a human learns to navigate obstacles independently, gradually understanding concepts like walls, cars, or animals as they mature. The flaw in current approaches is evident from recent incidents where vehicles misinterpret objects, mistaking a sunset for a red light, triggering sudden braking manoeuvres.

What progress has Minus Zero made thus far?

In 2021, we became the first company to conduct autonomous driving trials on public roads in India. Two years later, we successfully deployed a fully autonomous pod in controlled environments due to regulatory constraints. We are collaborating with multiple automakers to install next-gen autonomy features in their entire portfolio of vehicles.

Does Minus Zero possess the necessary computing and hardware capabilities?

Our expertise lies predominantly in AI. We possess the computational prowess to train AI models efficiently.

Moreover, we are ensuring that our algorithms remain energy-efficient, suitable for edge computing. Partnerships with leading Silicon-based companies and institutions ensure access to cutting-edge GPUs, addressing hardware challenges.

Does this mean Minus Zero only provides software solutions?

As a Tier-1 supplier to automotive OEMs, we offer comprehensive end-to-end solutions, encompassing sensors, computing hardware, and software. Collaborations with sensor and computing hardware providers complement our offerings, ensuring seamless integration into vehicles. In this ecosystem, Minus Zero serves as the primary interface to automakers, while chip manufacturers operate at a secondary level.

When can we expect Minus Zero-powered vehicles on the roads?

Expect announcements regarding our partnerships with automakers within the next 3 to 4 months. While it may take from 1 to 2 years for the product launch and testing a substantial number of vehicles on the real world.

What testing methodologies has Minus Zero employed thus far?

Testing is an ongoing process, spanning multiple stages. We have conducted software-in-loop, hardware-in-loop, and vehicle-in-loop tests using simulators, campus environments, and real-world scenarios.

Will Minus Zero focus solely on electric vehicles (EVs)?

Our technology transcends vehicle types. Any vehicle equipped with necessary interface can integrate Minus Zero technology seamlessly, enhancing autonomy beyond conventional bounds.

Which vehicle segment do you anticipate adopting this technology more enthusiastically: commercial or personal vehicles?

Both segments present unique challenges, and our solution caters to both segments. In the commercial vehicle (CV) realm, our system acts as a co-pilot, streamlining operations. It's equally invaluable in environments like ports and airports, facilitating continuous material and human movement. Personal vehicle (PV) adoption has already gained momentum, with more focus on customisations as per individuals' needs.

What will be the inaugural product under the Minus Zero brand?

Our inaugural product targets highway autonomy, offering consumers hands-free eyes-on driving experiences upon entering highways. Subsequently, we will introduce solutions tailored for urban roads as well.

Why prioritise highway autonomy initially?

With the mobility market ripe for disruption, our phased approach commences with highway autonomy. Highways contribute to the highest number of road accidents; hence with our ADAS features, we'll not only ensure safety but also add to comfort features for drivers over long stretches of road. Urban road solutions will follow, leading to comprehensive autonomy, thereby accommodating varying adoption rates.

In the long run, how will Minus Zero generate revenue?

Our revenue model will evolve continually, aligning with our vision of establishing our solution as the quintessential autonomy platform for vehicles. As a Tier-1 supplier to automotive OEMs, our initial primary revenue stream would be generated from selling our AI platform for autonomy beyond ADAS. Whereas, subscription-based services, leveraging the wealth of data generated, will also be pivotal in areas like insurance and resale, constituting a significant aspect of the software-defined vehicle architecture. **EFY**